

A 12V/24V-to-1V Shared Switched-Capacitor Multi-Inductor Multi-Output Converter with 90.9%/89.5% Peak Efficiency and Negligible Cross Regulation

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In modern electronic systems like datacenter, automotive applications, etc., multiple low-voltage power rails are essential for high power efficiency and fine-grained power management. To minimize I²R losses in the power delivery network, a higher input voltage is preferred. As a result, a high-efficiency high-density power management unit with high V_{IN} and low voltage conversion ratio (VCR, defined by V_{OUT}/V_{IN}), along with multiple outputs, is highly demanded.

As shown in Fig. 1 (top left), compared to the two-stage conversion with cascaded efficiency loss, direct 12/24/48V to 1V power delivery with high efficiency has become increasingly popular. To improve the power conversion efficiency under low VCR, many hybrid step-down topologies, by stacking low-voltage power switches with a better figure-of-merit (FoM) and flying capacitors, have been proposed [1-5]. Recently, several hybrid converters with multiple outputs have been proposed for further improving the power density [6-9]. A three-level conversion or series-capacitor conversion front stage [6], [7], allows capacitors and inductors to be shared among those outputs. However, extra power switches on high-current paths are needed to direct inductor current to those independent outputs, degrading efficiency. Pioneering works that use one inductor for one output and share flying capacitors [8] or DC capacitors [9] reduce component counts while achieving high efficiency and low output ripple. In [8], this shared series-capacitor structure achieves three outputs with fewer components, where the switches face high voltage stress, thus high-voltage devices are required. Moreover, shared flying capacitors require charge balancing, making the output voltages of the three channels dependent. In [9], a ladder structure reduces the switch voltage to $V_{IN}/3$, enabling the use of 5V devices and easy system expansion to support more outputs with shared DC capacitors. Nevertheless, each output still requires its own set of flying capacitors to ensure charge balance, resulting in more off-chip components under more outputs. Additionally, the fixed switch-rated voltage ($V_{IN}/3$) prevents the structure from being directly extended to handle higher input voltages, such as 24V.

In this work, we propose a multi-output low-VCR DC-DC converter that not only shares the flying capacitors in each hybrid cell but also allows each hybrid cell to operate independently, as shown in the bottom right of Fig. 1. Each output hybrid cell is connected in series to reduce the voltage stress on switches and inductors. This connection approach reduces external components, enhancing the ability to support higher input voltages with more output channels (N). With all switches experiencing voltage stress of $V_{IN}/(N+1)$, better FoM 5V devices can be used, enhancing system efficiency. To address the issue of non-independent output power across channels caused by the shared flying capacitor [8], we introduce a charge redistribution (CR) block. Thus, the duty cycle of each channel D_i is only determined by the input voltage (V_{IN}) and its output voltage (V_{O_i}), expressed as $D_i = (N+1) \times V_{O_i}/V_{IN}$.

As illustrated in Fig. 2 (top left), the hybrid cell of the proposed multiple-output converter consists of three switches M_{Ai} – M_{Ci} , a flying capacitor C_{Fi} , a DC capacitor C_{DCi} , and an inductor L_i . During the magnetization period Φ_1 of L_i , M_{Ai} is turned on, L_i and C_{Fi} are charged from the up-side hybrid cell. During the inductor demagnetization period Φ_2 , C_{Fi} transfers the stored charge in phase Φ_1 through the switches M_{Bi} and M_{Ci} to the down-side hybrid cell. This operation like [8] will make each output power correlated because the charge extracted by output $i+1$ will be equally transferred to output i . The introduced CR block effectively solves this issue by using a DC capacitor (C_{CP}) as a charge pool to redistribute residual charge at V_R through switches M_{Ri} . Thus, at the output side, each hybrid cell can be paralleled to provide multiple independent outputs. Specifically, during Φ_1 , an auxiliary path through M_{Ri} is added to decouple the output current (I_{O_i}) and the current through the flying capacitor C_{Fi} (I_{CFi}). If the load at output- i is much heavier than the other channels, M_{Ri} can recharge the output- i (I_{Ri}). Thus, C_{Fi} transfers less charge to the down-side cell with light load and requires less charge from the up-side cell, leading to the charge balance of all the flying capacitors. Otherwise, the recharge current (I_{Ri}) can also be negative when cell- i has a lighter load. By sequentially connecting the UP of each of the N hybrid cells to the DOWN of the next, and connecting the DOWN of the last hybrid cell to a C_{CP} which then feeds back to each cell's V_R , an $(N+1):1$ switched-capacitor (SC), N -output DC-DC converter can be implemented. Under the two-phase operation, the voltages across C_{DCi} (V_{DCi}), C_{Fi} (V_{Fi}) are then derived as $(i+1) \times V_{IN}/(N+1)$ and $i \times V_{IN}/(N+1)$, respectively. As a result, all the switches have a voltage rating of $V_{IN}/(N+1)$.

Fig. 2 takes a dual-output converter which consists of two basic cells as an example to illustrate the operating principle of the proposed scalable topology. The top right of Fig. 2 presents the calculated ratio between I_{Ri} and I_{CFi} under varying load conditions. When I_{O2} is two times higher than I_{O1} , during Φ_1 , C_{F1} is only charged by I_{O1} through L_1 . To avoid excessive charge transferred to cell-1, I_{R2} will also provide charge to I_{O2} during Φ_1 . As for channel-1, since I_{O1} is smaller, during phase Φ_1 , C_{F1} transfers the extra charge to C_{CP} or cell-2 through M_{R1} . With the CR block, each output current remains unaffected by others—even when one output is unloaded, its capacitor network can be reused by a fully loaded path, significantly enhancing component utilization. The C_{DCi} in each cell allow the duty cycle overlap, enabling simultaneous rise of all inductor currents. The independent transient responses of each channel ensure fast transient response.

Fig. 3 left presents the circuit block diagram of the proposed 3:1 SC dual-output converter, comprising two power-stage cells, each integrating the power MOSFETs and controller. Except for M_{A2} , all gate drivers can directly draw current from existing DC voltages on C_{DC1} – C_{DC4} and C_{CP} , significantly reducing the required bootstrap capacitance and conserving chip area. The drivers for M_{C1} and M_{C2} are powered directly by V_R , while those for M_{C1} and M_{C2} are powered by V_{DC1} and V_{DC2} , respectively. As there is no voltage higher than V_{IN} , only M_{A2} requires a bootstrap capacitor, C_{BST} . A specialized driver, illustrated in the top-right of Fig. 3, is designed for M_{R1} , M_{R2} , and M_{A1} . This driver allows these transistors to be turned on using V_{DC1} or V_{DC2} and to return charge to their source when turning off, eliminating additional switching losses. The V_{FLOAT} which is slightly lower than $V_{IN}/(N+1)$ is only used during the start-up process when the DC voltages have not been established. Furthermore, independent peak-current mode control with slope compensation is employed to generate pulse-width modulation signals for each output, simplifying the interconnection between chips. The chips can be easily connected in series, as illustrated in Fig. 3 (right bottom), where the source of M_{C1} in chip2 is externally connected to the drain of M_{A2} in chip1, and both chips share a common C_{CP} . This configuration enables a 5:1 SC quad-output converter supporting a 24V input voltage, realized with two 3:1 dual-output chips, with all power switches implemented using 5V devices.

We implemented a dual-output converter in a 0.18 μ m BCD technology, operating at 2MHz. Each output utilizes a 470nH inductor with 21m Ω DCR (0806 package) and three 10 μ F capacitors (output, DC, and flying, all 0805), delivering a peak current of 4A. Meanwhile, two dual-output converters are used to implement a quad-output converter with a total volume of 50.3 mm³. Fig. 4 shows the measured startup and steady waveforms of the quad-output converter. As input voltage reaches 24V, input decoupling capacitor C_{DC4} holds V_{IN} , while the voltages across C_{DC3} , C_{DC2} , C_{DC1} , and C_{CP} eventually stabilize at $4 \times V_{IN}/5$, $3 \times V_{IN}/5$, $2 \times V_{IN}/5$, and $V_{IN}/5$, respectively. Fig. 4 also shows the load transient response of the dual-output converter. When channel-2 operates at full load, channel-1 undergoes a 4A current step with a 1.2 μ s rise time, resulting in a 170mV undershoot. Similarly, with channel-2 at light-load, channel-1 experiences the same current step, with a 166mV undershoot. In both cases, no cross-regulation is observed, validating the independence of the outputs. With continuous inductor current to the output, an output voltage ripple of about 10mV is achieved.

Fig. 5 presents the measured efficiencies under dual- and quad-output conditions. When $V_{IN}=12V$, the proposed dual-output converter achieves a peak efficiency of 92.6% at $V_{O1}=V_{O2}=1.8V$ and $I_{O1}=I_{O2}=0.9A$. When $V_{IN}=24V$, the quad-output converter reaches a peak efficiency of 91.1% at $V_{O1}=V_{O2}=V_{O3}=V_{O4}=1.8V$ and $I_{O1}=I_{O2}=I_{O3}=I_{O4}=1A$. With the shared switched capacitors, the measured peak efficiency drops by only 1.5% when the input voltage is doubled, which indicates that the advantages of the proposed topology become more pronounced as the number of outputs increases. For unbalanced load conditions, the dual-output converter also achieves high efficiency. With $V_{O1}=V_{O2}=1.2V$ and $I_{O1}=0A$, a peak efficiency of 88% is measured at $I_{O2}=1.1A$. Similar peak efficiency is measured when $I_{O2}=0A$ and I_{O1} varies.

Fig. 6 presents the performance summary and comparison with the state-of-the-art designs. Leveraging the shared SC topology, this quad-output converter with only two capacitors per output achieves 3.5% higher efficiency than that of [1] with four capacitors. To our best knowledge, this is the highest power efficiency for published multi-output 24V-to-1V converters. With quad-output, the chip area of the proposed converter is similar to [9]. It not only reduces the number of capacitors by two but also operates at nearly twice V_{IN} , while still achieving a peak efficiency of 89.5% at VCR = 0.05 for $V_{O1-4}=1.2V$, and $I_{O1-4}=1A$, comparable to [9] at VCR $\approx$ 0.1. Since switch voltage stress in [9] is $V_{IN}/3$, supporting a 24V input would require devices with at least 8V ratings. In contrast, our quad-output converter, with a switch voltage stress of $V_{IN}/5$, only requires 5V device. And also, this topology has the potential to scale to 48:1 multi-output application. Fig. 7 shows the die micrograph of the converter alongside the PCB implementation.

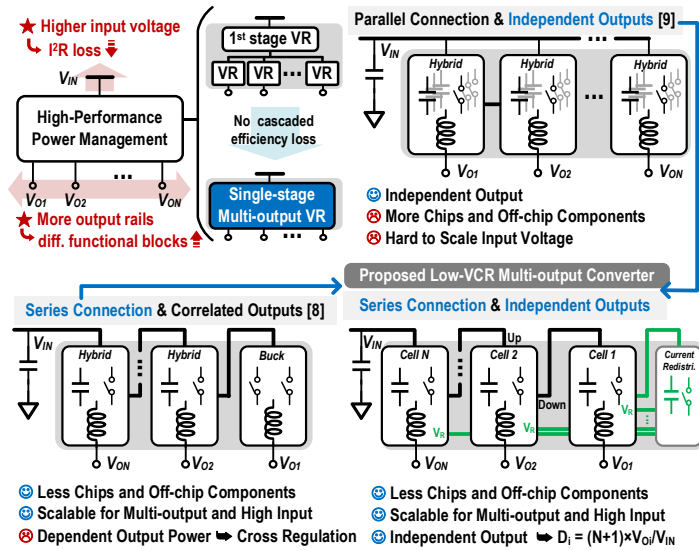


Fig. 1. Conceptual diagrams of the prior and proposed low-VCR multi-output buck converters.

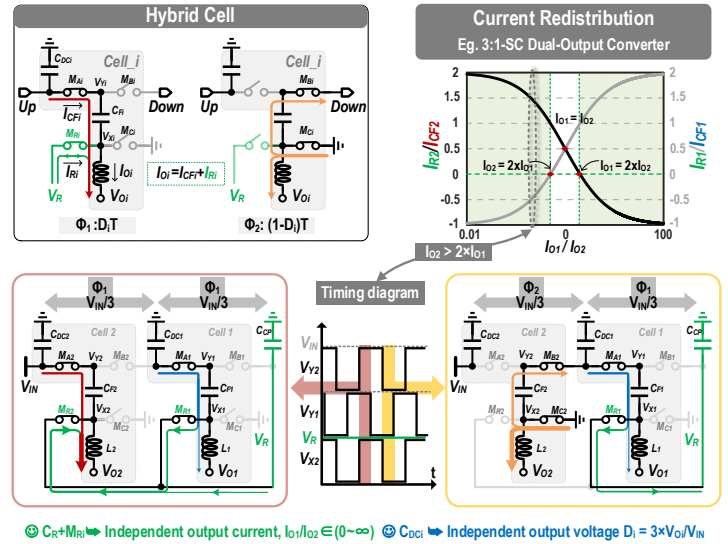


Fig. 2. Proposed hybrid cell and working principle of the proposed 3:1-SC dual-output converter

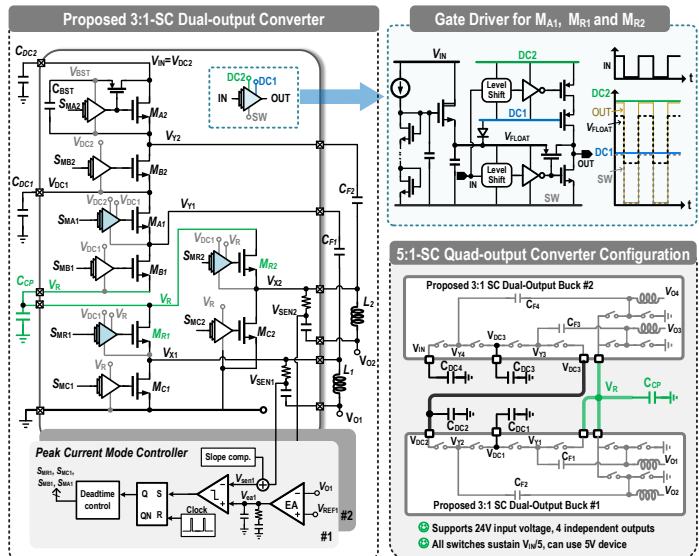


Fig. 3. Simplified diagram of the proposed 3:1-SC dual-output converter, and the implementation of 5:1 SC quad-output converter.

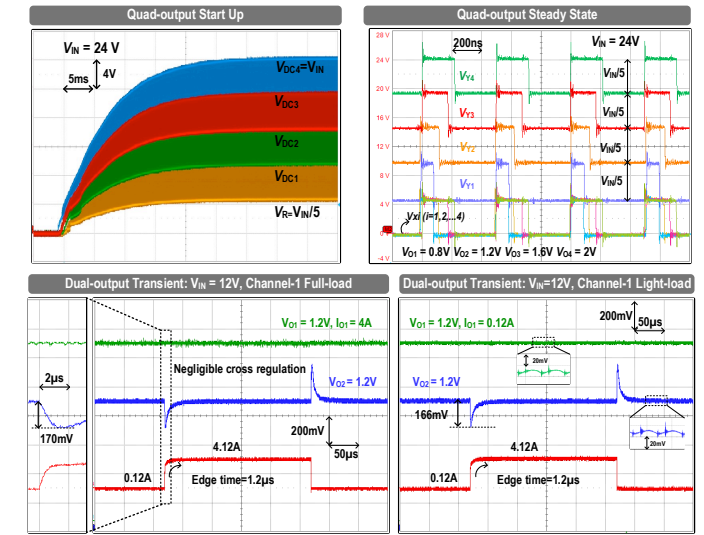


Fig. 4. Measured start-up, steady waveforms (top) of the proposed 5:1-SC quad-output converter, and load transient waveforms (bottom) of the 3:1-SC dual-output converter.

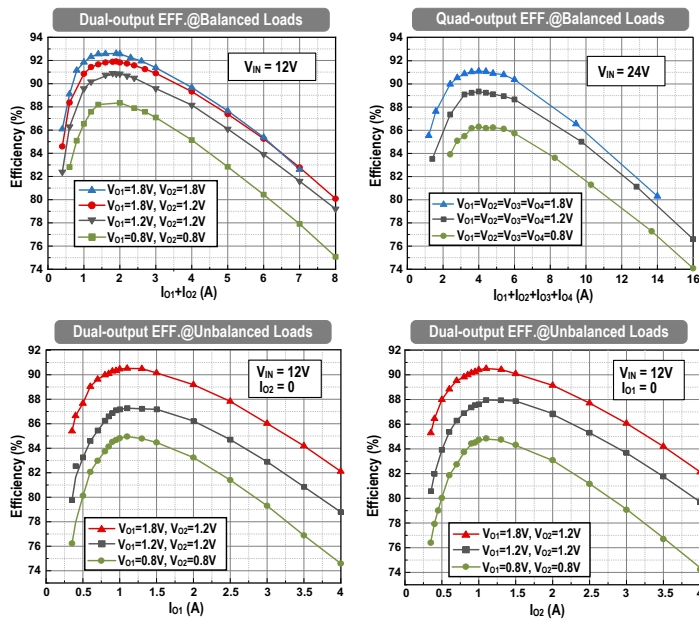
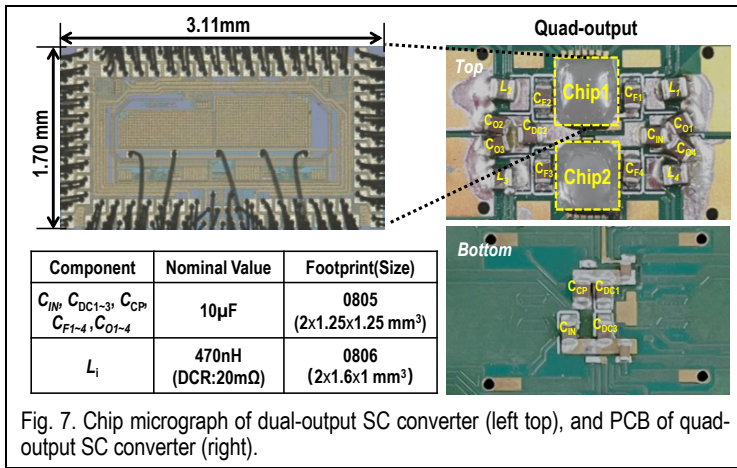


Fig. 5. Measured power conversion efficiencies under dual- and quad-output configurations.

	This work	[9] ISSCC '23	[8] APEC '23	[7] ISSCC '23	[1] ISSCC '24
Process(μm)	0.18 BCD	0.18 BCD	Discrete	0.15 BCD	0.18 SOI
Chip Area(mm ²)	5.29	2×5.29	4×2.5	7.29	10.26
V _{IN}	12V	24V	12V	24V	48V
V _{OUT}	0.8–1.8V	0.8–1.8V	0.8–1.8V	1.2–2.2V	1.3–1.8V
Switching Frequency (Hz)	2M	2M	2M	0.3M	2M
L (H) & DCR (mΩ)	470n & 20	470n & 20	220n & 6	400n & 0.18	1μ & N.A
@package(mm ³)	2×1.6×1.2	2×1.6×1.2	3.2×2.5×1.2	10×7×9	N.A
C _O (μF)	10	10	22	20	10
C _F (μF)	10	10	4.7	2.8	1
Number of Outputs	2	4	4	3	2
Number of Cap. for each output	2	2	2.5	0.7	0.5
Max. Switch Voltage	V _{IN} /3	V _{IN} /5	V _{IN} /3	V _{IN}	V _{IN}
I _{MAX} (A)	8	16	16	27	4
Ripple (mV)	<10	<10	N.A	N.A	25
Droop (mV/A)	42	42	30	N.A	54
Cross-Regulation (mV/A)	0	0	0	N.A	25
Peak Efficiency @1/VCR	90.9% @ 10	89.5% @ 20	89.8% @ 10.9	91.8% @ 16	77% @ 30
					86% @ 20

*: Estimated from graph.

Fig. 6. Performance summary and comparison with state-of-the-arts.



Acknowledgements

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